Paleomagnetism of the Circum-Ungava Proterozoic Fold Belt (I): Cape Smith Komatiitic Basalts

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A total of 397 oriented cores were taken from pillowed, columnar, and massive basalt of Cape Smith Island in northeastern Hudson Bay and from the adjacent mainland coast of Quebec. The thick series of basalts form part of the Circum-Ungava Proterozoic Belt which is of Aphebian (early Proterozoic) age. The lower portion of the basalt series contains pyrrhotite (Fe_7S_8) and did not yield useful paleomagnetic results. The magnetization of the upper portion of the series is carried by small quantities ($\ll$1 wt %) of magnetite. The directions of magnetization of the samples do not change much during alternating field and thermal washing, and show very good grouping. The available internal evidence on the age of the stable remanence is inconclusive. The pole position (20°N, 162°E, semi-axes of the 95% oval of confidence: 4° and 5°) after correcting for tilt is about 70° west of the preHudsonian part of the polar wandering curve established from North American data. The uncorrected pole position (16°N, 107°W), which would apply if the magnetization was acquired at the time of the Hudsonian orogeny, is in the general area of paleomagnetic poles for rocks of similar age from other localities of the Canadian Shield.

Un ensemble de 397 carottes orientées furent prises dans des basaltes massifs, en oreillers et en colonnes de l'île du Cap Smith située dans la partie nord-est de la Baie d'Hudson, ainsi que dans la région côtière adjacente du Québec. Les séries épaisses de basaltes font partie de la ceinture Circum-Ungava d'âge Aphébien (Protérozoique inférieur). La partie inférieure des séries de basalte contient de la pyrrhotine (Fe_7S_8) et ne produit pas d'utiles résultats paléomagnétiques. La magnétisation de la partie supérieure des séries est portée par de faibles quantités ($\ll$1% pd.) de magnétite. Les directions de la magnétisation des échantillons ne changent pas beaucoup pendant le lessivage thermique et de champ alternatif, et montrent un très bon regroupement. Les évidences internes disponibles sur l'âge de la rémanance stable sont peu concluantes.

La position du pôle (20°N, 162°E, demi-axes du niveau de confidence 95%: 4° et 5°), après corrections pour l'inclinaison, est à environ 70° à l'ouest de la partie pré-hudsonienne de la courbe de déviation polaire établie à partir des données nord-américaines. La position non corrigée du pôle (16°N, 107°W), que l'on appliquerait si la magnétisation avait été acquise au moment de l'orogénie Hudsonienne, est située dans la région générale des pôles paléomagnétiques des roches d'âge semblable provenant d'autres localités du Bouclier Canadien.

[Traduit par le journal]

Introduction

The present plate boundaries are primarily defined by zones of high seismicity. However, in the search for ancient plate boundaries or plate interaction zones, paleomagnetism must play a leading role. Paleomagnetism has also been very useful in tracing the evolution of the present plate tectonic system.

The Canadian Shield has been divided into a number of structural provinces on the basis of predominant ages and orogenies (Stockwell 1964). In the last few years, paleomagnetism has been used to determine whether at least some of these provinces were separate plates or parts of plates that came together along subduction zones like India and Asia during the Mesozoic. The method consists of a search for abrupt changes in the paleomagnetic polar wandering pattern from one province to another for certain time intervals. For instance, the paleomagnetic poles for the Grenville province have been compared to results from elsewhere in North America for the same time interval. The results of these studies imply that the Grenville province has moved relative to the older Precambrian provinces, but the characteristics of the motion are uncertain.

North of the Grenville Province, an arcuate structure composed mainly of early-Aphebian sedimentary and basaltic rocks belonging to the Churchill Province (main orogeny dated at 1750 Ma (million years)) is in contact with rocks belonging to the Superior Province (main orogeny dated at 2500 Ma). As shown on Fig. 1,

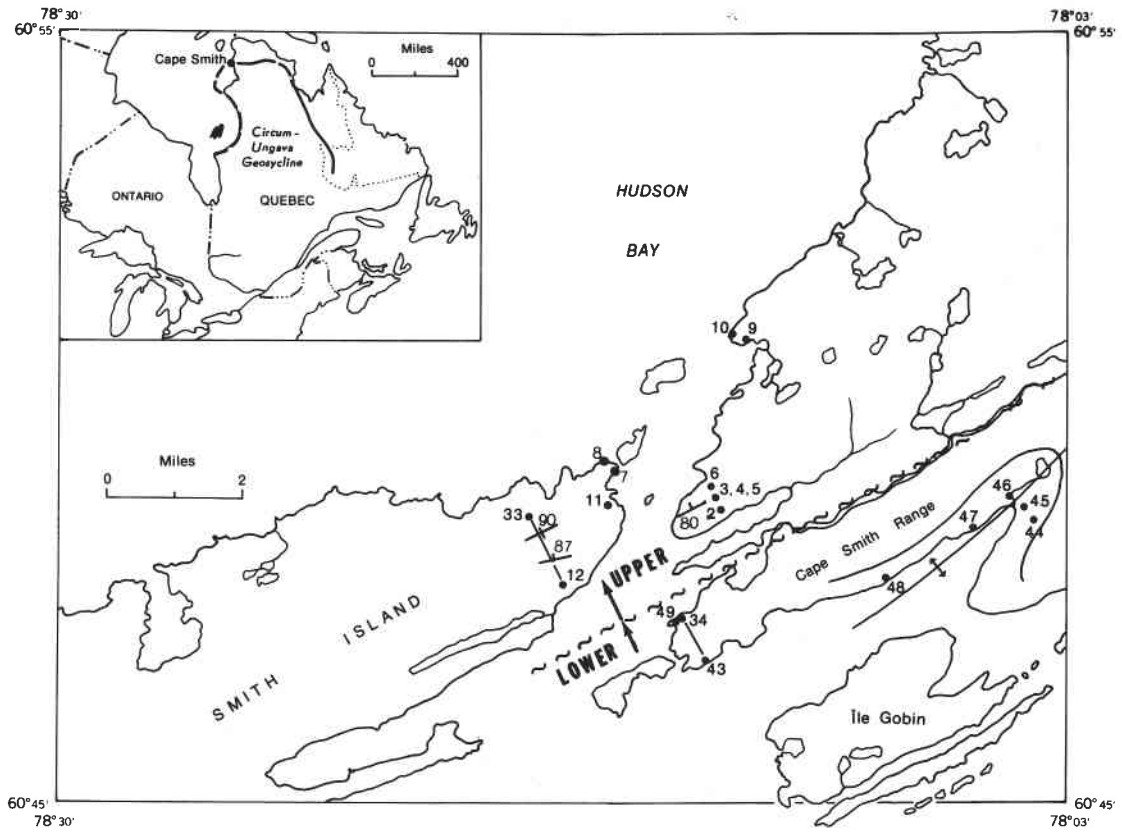


FIG. 1. Location of sampled area and of the collection sites.

this structure runs from South of the Grenville–Superior boundary via Schefferville (Quebec) to the West coast of Ungava Bay, then westward across Northern Quebec to the East coast of Hudson Bay (Cape Smith), where it bends southwards and essentially follows the coast line of Hudson Bay east of the Belcher Islands. This arcuate structure, formed by rocks amenable to paleomagnetic study, offers another approach in detecting ancient movements between the Churchill and Superior Provinces. The test is based on a comparison of directions of stable remanent magnetization along this 2500 km-long, arcuate structure, to see whether its shape is primary or the result of deformation. Similar pole positions would be expected for various parts of the belt if the structure is primary. The first collection of samples in this study was carried out on Smith Island and on the mainland (New Quebec) just to the East of the Island. A further application of the paleomagnetic data will be in stratigraphic correlation of lithologic units from different segments of the belt.

Geology

The geology of the Cape Smith portion of the Circum-Ungava belt has been described by, among others, Stevenson (1968), Stam (1961), Bergeron (1957), and Taylor (1975). Dimroth *et al.* (1970) have compiled and analyzed the geological data for the whole belt.

The Proterozoic rocks in the Cape Smith belt are usually subdivided into a lower series, consisting predominantly of both clastic and volcanogenic sediments, and an upper series, consisting predominantly of basaltic flows and some sills. It is not clear whether an unconformity occurs between the series (Taylor 1975). A number of thrust planes have been observed in the lower series (Stevenson 1968). Greenschist facies metamorphism is reported for the eastern part of the belt, but in the western part (Cape

Smith Island), primary minerals are generally present and the metamorphic grade is evidently sub-greenschist facies.

Sampling was confined to the almost vertically dipping upper series in the extreme western part of the belt. Total thickness of the basalts is about 6000 m. Pillows are abundant; they show no obvious signs of deformation and indicate tops to the northwest. The central portion of many pillows is composed of parallel layers of quartz and calcite, separated by thin layers of basalt. One explanation for this unusual feature is that the pillows leaked intermittently while solidifying. Columnar jointing was noticed in places as well as massive flows or sills. The rock has a fresh appearance, with minor chloritization. A broad fold was observed in the sampled area.

A total of 391 drill cores of diameter 2.5 cm was collected from the basalts of the upper series, and 6 cores from a conformable layer of ultrabasic rock. Localities are shown on Fig. 1. A sun compass was used in orienting almost all of the cores. Samples were collected from the limbs of one fold in the lower part of the sampled sequence, to test the authenticity of the remanent magnetization. A site is an outcrop or selected small area from which at least six cores were taken.

Mineralogy and Chemistry

Twenty samples from various horizons of the stratigraphic column were analyzed with a recording thermomagnetic balance. All samples from the lower portion of the basalt sequence show saturation magnetization versus temperature curves with two Curie points at 300–310 °C and at 540–580 °C as illustrated on Fig. 2A. The samples from the upper portion yield only one Curie temperature in the range of 550–580 °C (Fig. 2B). Examination with a microscope revealed the presence of pyrrhotite in the samples from the lower portion of the series. Thus, the Curie temperature of 300–310 °C is ascribed to the magnetic pyrrhotite type Fe_7S_8 (Schwarz 1974). The high Curie temperature mineral occurs in very small quantities (well below 1% by weight) throughout the basalt series. This mineral is either an oxidized titanomagnetite (titanomaghemite), which is known to be formed by slow oxidation of primary titanomagnetite in pillowed oceanic basalts, or almost pure magnetite formed by exsolution of Ti-rich oxides. The variation in Curie temperature in the series is small and not systematic. The magnetic content is rather lower than that generally expected for basalts. However, the sampled ultrabasic layer contains as much as 1% Fe_3O_4 by weight, in the form of small grains. Optical examination of the rock suggests that the relatively high magnetic content may be due to serpentinization, producing secondary magnetite.

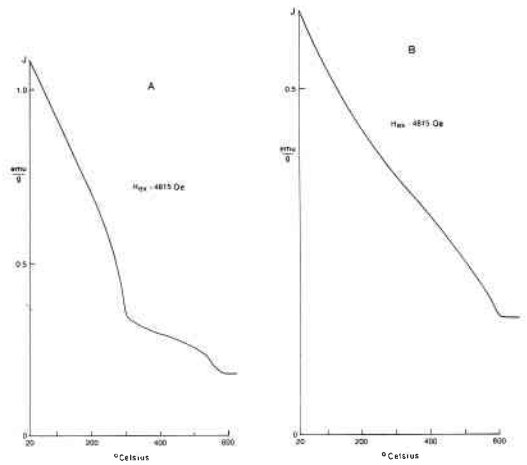


FIG. 2. Typical thermomagnetic curves for lower portion of basalt series (A) and upper portion (B).

The silicates are clinopyroxene and plagioclase of intermediate composition. Secondary minerals are chlorite, sericite, actinolite and serpentine (olivine alteration), and sphene. Silicate alteration is extensive in many samples. Chemical analyses for 12 samples show that (1) total Fe content corresponds to 10 to 15 weight % Fe_2O_3, and (2) FeO/Fe_2O_3 is between 4 and 6. These values are comparable with those reported by Irving *et al.* (1970) for basalt samples from the Mid-Atlantic Ridge. TiO_2 content is about 0.5%. An interesting aspect is the high MgO content (up to 24 weight %) of all basalt samples used in the analyses. As shown on Fig. 3, the samples fall in the tholeiite field as defined by Kuno (1950), rather than in the calc-alkaline field. Three out of twelve samples meet the requirements of Brooks and Hart (1974) for classification as komatiites. A report on the chemistry and textures in these basalts is in preparation.

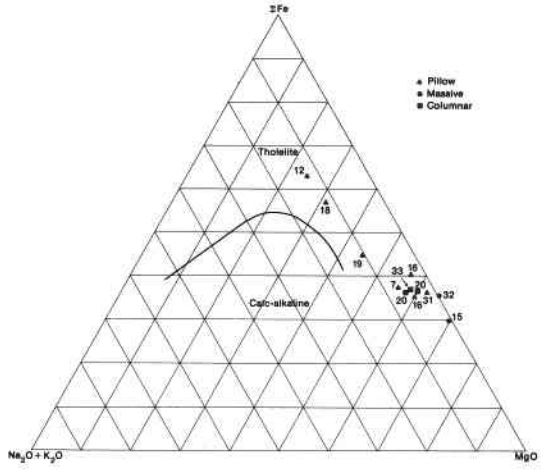


FIG. 3. Chemical composition of samples from sites as numbered.

Remanent Magnetization

Two cylindrical specimens (2.5 cm in diameter, 2.2 cm long) from each drill core were used for remanence measurement and partial demagnetization in alternating magnetic fields (AF) of up to 800 Oe. The latter was done to remove recently acquired, low stability components of remanent magnetization, with the objective to isolate original remanent magnetization acquired during cooling (thermoremanent magnetization, TRM) and/or possibly shortly thereafter by oxidation of original titanomagnetite or production of secondary magnetite during serpentinization (chemical remanent magnetization, CRM). This original remanent magnetization is expected to show good stability; that is, sufficiently high coercive force. A third specimen from 40 cores was thermally demagnetized.

The magnetization direction of each core was required to show reasonable homogeneity during AF demagnetization. For that reason, the results for a core were rejected if the directions of its specimens differed by more than 25°.

The Natural Remanent Magnetization (NRM) is relatively weak for these basalts (Table 1). The NRM directions of the lower part of the sampled sequence are scattered and have a low degree of within-core consistency. During stepwise AF demagnetization, the intensity decreased sharply in the initial stages and showed irregular variations in higher AF fields. Grouping of the core directions and the within-core consistency did not improve using fields up to 800 Oe, and consequently the results for the lower part were discarded in this study.

The remanent intensity decreased slowly during AF demagnetization which indicates good stability for the upper part of the sampled sequence. The changes in direction during AF demagnetization are consistent and usually small as seen on Table 1 and Fig. 5. Table 1 shows that the results for the samples collected on the mainland (Quebec) and those taken from Smith Island do not differ appreciably. In general, the site mean directions become better defined because of better grouping of the within-site directions.

Thermal demagnetization of cores from the lower part of the sequence also failed to isolate a stable NRM component. No end points of the magnetic vector could be defined, on account of irregular rotations. The results of a storage test for these cores, which extended over 5 months, shows rotations of 10° to 160°, indicating the viscous nature of a substantial part of the remanent magnetization. This explains the scatter in NRM directions, and may have been a factor in the failure of isolating stable remanence. The large proportion of viscous remanent magnetization in these samples is probably due to the presence of disseminated, fine-grained Fe_7S_8.

On the other hand, thermal demagnetization of 10 samples from the upper part of the sampled sequence yielded results (Fig. 6A) that are comparable to those obtained by AF demagnetization, as shown on Fig. 5. The mode of decay of the remanent intensity suggests that samples from massive layers have unblocking temperatures distributed in a lower temperature range than samples from pillowed and columnar layers (Fig. 6B, C, and D). The magnetic mineralogy being constant as suggested by the thermomagnetic curves, the grain size of the remanence carriers may well be somewhat smaller in at least the rapidly cooled pillowed basalts.

Magnetic Fabric

The directions of the principal susceptibilities were determined for 80 specimens (40 cores) in an attempt to obtain more information on the authenticity of the magnetization. The directions for the maximal (k_{max}) and minimal (k_{min})

TABLE 1. Natural Remanent Magnetization values for oriented basalt samples

Site No.	N	Rock Type	NRM I ($\times 10^{-6}$)	NRM D°	NRM I°	NRM σ°	NRM K	NRM α95°	After AF-demagnetization H (Oe)	After AF-demagnetization I ($\times 10^{-6}$)	After AF-demagnetization D°	After AF-demagnetization I°	After AF-demagnetization σ°	After AF-demagnetization K	After AF-demagnetization α95°	Pole Position
									Mainland (Quebec)							
1	7	a	1.55	273	57	31	7	20	200	0.68	242	54	11	52	9	
2	6	a	2.88	239	77	10	68	7	200	2.22	212	64	7	151	5	
3	6	b	4.63	233	72	24	12	17	200	2.63	227	59	10	67	7	
4	5	b	3.52	209	74	26	10	18	200	2.33	220	66	30	7	24	
5	4	b	1.97	191	81	49	3	38	200	1.68	216	57	12	44	11	Lat. and Long. (semi-axes of 95% confidence area) [after tectonic correction]
6	5	a	3.70	229	71	39	4	30	200	1.13	219	64	22	13	17	
7	8	b	1.62	160	58	70	1	43	250	0.05	223	51	21	15	13	
8	7	b	0.16	204	67	9	90	6	250	0.07	204	60	7	152	4	
9	9	b	2.97	223	67	19	17	11	300	1.46	234	53	17	22	11	
10	9	a, b	6.91	291	63	24	11	14	300	3.03	303	45	22	14	16	
11	6	b	1.59	248	55	14	36	10	200	0.74	239	47	19	18	13	
									Mean		231	60	17	24	9	19N, 117W (7, 5) [18N, 163E]
									Smith Island							
12	6	b	4.48	209	64	19	19	13	150	3.98	217	60	15	31	11	
13	5	a	4.40	209	54	12	49	9	250	1.68	202	51	13	40	10	
14	6	d	405.00	175	68	58	2	41	250	3.78	202	50	50	3	38	
15	6	c	75.0	229	76	58	2	40	250	9.53	192	55	25	10	18	
16	6	b	5.96	235	55	36	5	25	50	5.88	221	54	25	11	19	
17	7	a	127.0	200	61	3	679	2								
18	6	b	27.1	206	62	9	84	6	150	18.3	205	56	9	84	6	
19	6	b	74.8	204	54	22	14	15	350	55.9	205	56	14	33	10	
20	6	a	5.77	206	62	24	11	17	100	3.97	228	60	20	17	14	
21	6	a	8.38	210	66	4	521	3								
22	6	c	2.04	196	60	7	154	5	150	1.83	203	64	6	165	5	
23	6	b	3.24	206	65	11	54	8	100	5.57	201	63	4	487	4	
24	6	a	6.76	178	65	44	3	31	200	3.36	211	63	6	196	4	
25	4	c	28.3	211	65	5	260	4								
26	4	b	44.8	204	56	66	1	58	200	3.97	202	63	8	111	7	
27	6	c	3.65	223	69	16	27	11	150	2.53	221	63	5	222	4	
28	6	b	18.2	223	70	21	14	15	150	6.64	225	59	13	41	9	
29	6	a	14.9	212	63	14	32	10	250	5.18	220	62	6	195	4	
30	6	c	145.0	77	−40	73	1	52	250	6.23	214	56	13	41	10	
31	4	b	4.82	224	57	64	2	55	200	2.22	216	51	18	20	14	
32	7	c	6.80	244	64	48	3	32	200	3.25	228	67	27	9	19	
33	6	a	2.43	216	71	14	36	10	200	2.36	210	61	6	165	5	
									Mean		211	60	7	131	3	15N, 102W (4, 3) [22N, 160E]
									Mean (all sites)		218	60	12	47	4	16N, 107W (5, 4) [20N, 162E]

N: Number of cores. I: Intensity of magnetization ($\times 10^{-6}$emu/cm^3). D, I: Declination and inclination. σ: Angular standard deviation. K: Precision estimate. α95: Angle of confidence. a: Pillow basalt. b: Massive basalt. c: Basalt with columnar jointing. d: Ultrabasic rock.

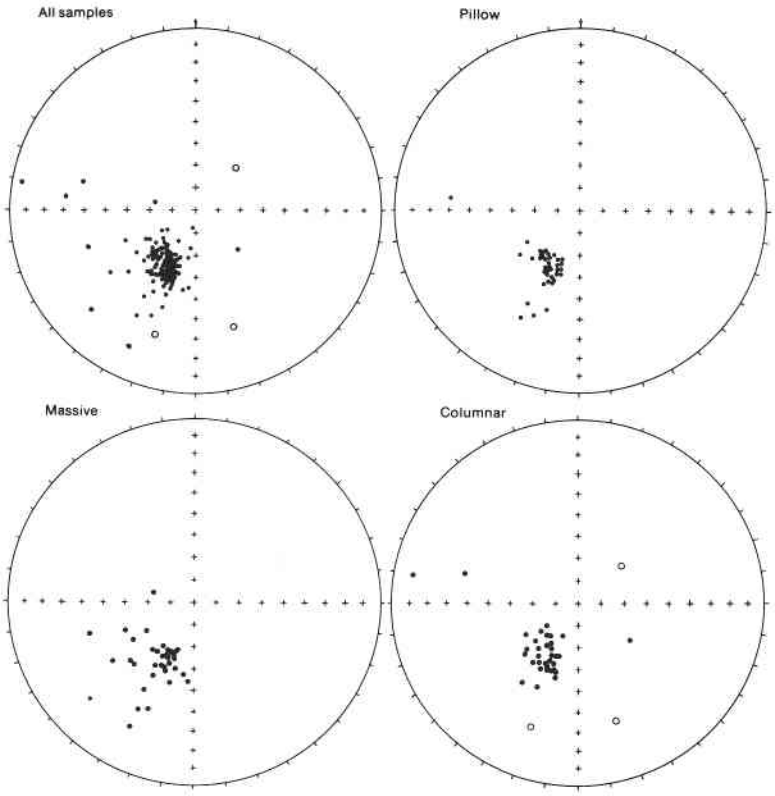


FIG. 4. Directions of natural remanent magnetization for the various types of basalt distinguished. Circles represent upper hemisphere projections.

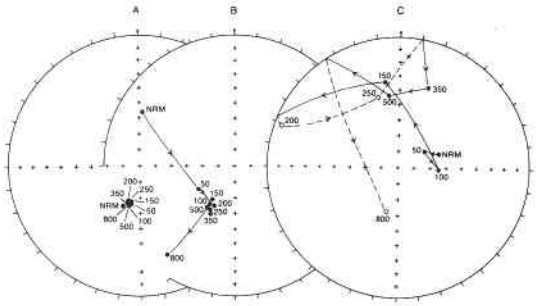


FIG. 5. Effect of alternating field washing on remanence directions for representative samples from upper portion (A and B) and lower portion (C) of sampled series.

susceptibility have been plotted on Fig. 7. The k_{max} directions, which probably represent the preferred orientation of the long axes of the magnetite grains in the samples, tend to lie in the bedding plane. The k_{min} directions are also strongly scattered, but nevertheless show a clear preference for a plane perpendicular to the bedding. These observations are in accord with both primary fabric due to flow and secondary fabric due to general tectonic compression directed NNW–SSE. Thus, the evidence is that the magnetic fabric is not younger than the folding (Hudsonian orogeny).

Paleomagnetic Pole

The stable directions obtained after demagnetization were averaged for each site and are listed in Table 1. These data were used in the calculation of a paleomagnetic pole position, assuming a geocentric dipole as the source of the field both before and after correction for the tilt of the sequence (see Table 1). Correction for the tilt could only be made by rotating about strike to the horizontal plane, because the geometry of the large folds is unknown. The results for Smith Island and the adjacent mainland combined were obtained giving all sites unit weight. The small limits of 95% confidence show that the pole position has been determined with high preci-

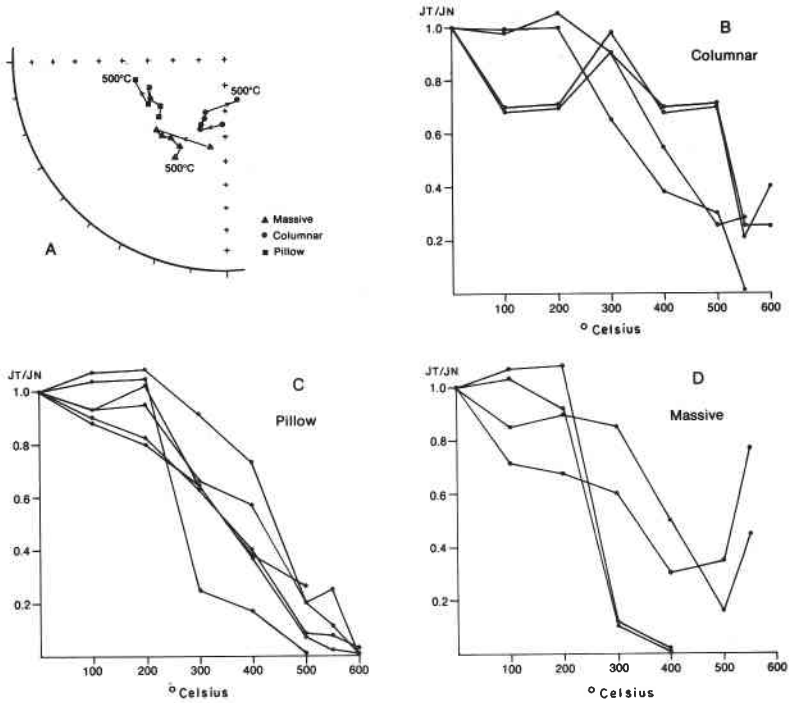


FIG. 6. Effect of thermal washing on directions (A) and intensity of remanent magnetization (B, C, and D) for representative samples. JT/JN indicates the ratio of the intensity after and before heating.

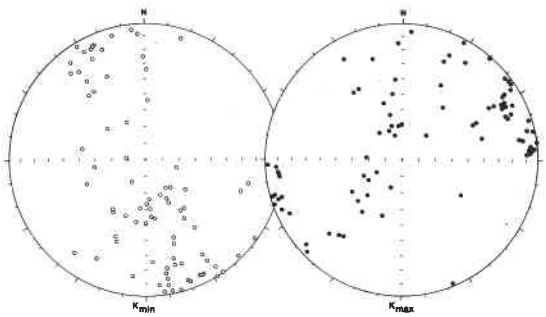


FIG. 7. Magnetic fabric expressed by preferred orientation of maximum and minimum susceptibility (k_{max} and k_{min} respectively) directions in sub-vertical ENE-striking basalt layers and pillows. Projection in lower-hemisphere only.

sion. In view of the large thickness of the series, the pole position is probably representative for a dipole field, the possible effect of secular variation has probably averaged out.

Age of Magnetization

The next question is whether this pole position represents the paleomagnetic field at the time of original cooling, or whether it reflects the field during a subsequent period of regional metamorphism such as the Hudsonian orogeny.

The following information is available:

1. In many samples, primary silicates show extensive alteration to low-grade minerals. Quench structures have been well preserved. Microscopic examination does not yield evidence for secondary opaques. Much of the original (Fe, Ti) oxides probably have been altered to sphene.

2. The K–Ar ages given by Beall *et al.* (1963) coincide reasonably well at 1450 to 1650 Ma. These ages are probably somewhat low because the rocks must be pre-Hudsonian. This suggests some loss of radiogenic argon during a low-grade metamorphic event.

3. The NRM directions of most of the samples change only slightly during demagnetization and they are well grouped. Both AF and thermal demagnetization have yielded conformable results and have isolated only one stable component of magnetization. These facts are compatible with stable remanent magnetization of single origin, in spite of the marked variation in mode of decay of remanent intensity. It is note-

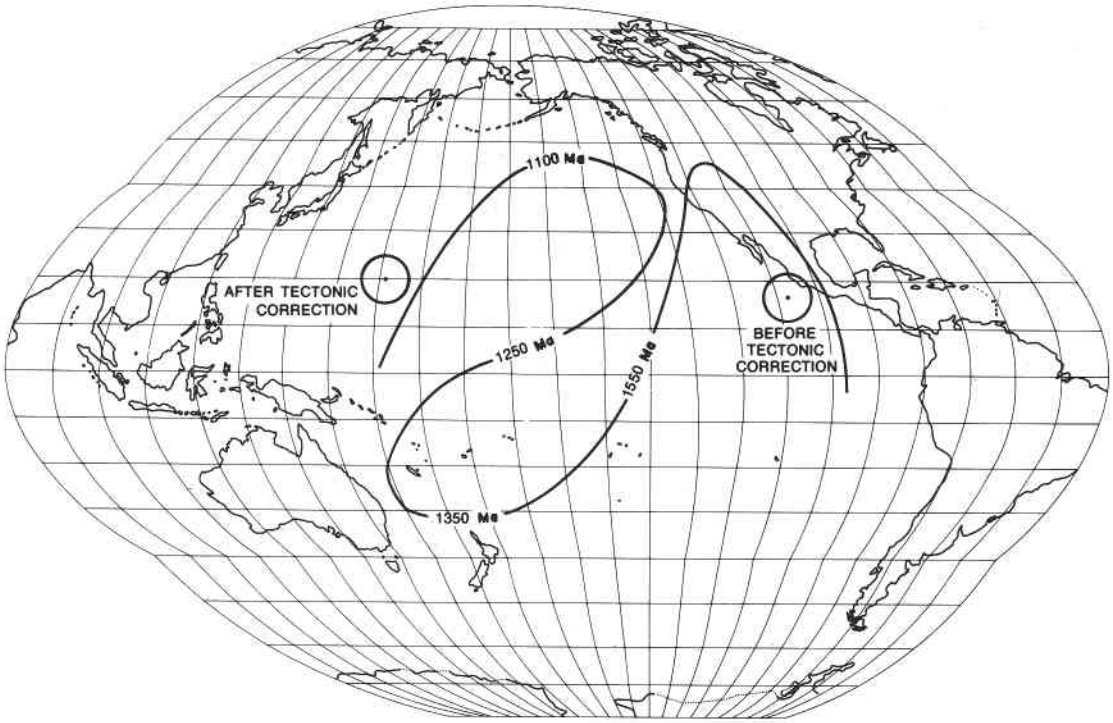


FIG. 8. Polar wandering curve for North American Precambrian (Fahrig *et al.* 1974) and pole positions before and after tilt correction for the Cape Smith basalts.

worthy that reversed directions of stable magnetization have not been found. Most intensity decay curves indicate good stability, but the unblocking temperatures are generally distributed between 200 and 550 °C. Thus, there is an obvious possibility that the magnetization was reset during a low temperature regional metamorphic event (see, for instance, the detailed discussion by Dunlop 1973).

4. Samples from pillows and columnar layers generally show a somewhat narrower blocking temperature spectrum, which is situated just below the Curie temperature (Fig. 6), than samples from the massive layers. This is compatible with somewhat smaller sized magnetic oxide grains in at least the rapidly cooled pillowed basalts. Also, the NRM directions of the samples from massive and columnar layers are more scattered (Fig. 4), suggesting the presence of low coercivity components that could reside in somewhat larger grains. This internal evidence is compatible with both an original magnetization and a secondary magnetization as mentioned under point 3, where stability is largely grain-size dependent.

5. Evidence from fabric studies indicates that the magnetic fabric is due to preferred orientation of magnetic grains, and is either primarily due to flow or is an effect of stress conditions existing during the Hudsonian orogeny. The strong scatter in principal directions may be related to complex flow patterns in pillows.

6. The ultrabasic layer is the only rock showing appreciable alteration. This is serpentinization, which usually generates secondary magnetite. In this case too, magnetite may well have been formed as shown by thermomagnetic results and high intensity of NRM (Table 1). However, serpentinization usually occurs shortly after emplacement of the rock, which means that the magnetizing field probably had not changed appreciably before magnetization of the secondary magnetite. Table 1 shows that the average direction for this layer ($\alpha_{95} = 38°$) falls within the group of basalt directions. Consequently, this fact cannot be taken as an indication that the stable remanence was acquired well after the formation of the series.

7. The NRM intensity is low compared to that for most basalts. This could be due to relatively low oxygen fugacity of the magma. However, it has become apparent that many

weakly metamorphosed rocks in Precambrian belts in Canada are weakly magnetized. At this time, the possibility cannot be discounted that primary magnetic material has been consumed in the formation of silicates such as sphene. In that case, however, the remanence direction would have remained unchanged.

8. Oxidation of low-Curie temperature titanomagnetites is known to occur in pillowed oceanic basalts (Schaeffer and Schwarz 1970; Ozima 1971). However, it seems that oxidation occurs either rapidly enough (Ozima *et al.* 1974) that the rocks retain their original remanence direction or that the secondary chemical remanent magnetization in some cases parallels the original thermoremanent magnetization (Marshall and Cox 1972). The Cape Smith pillows show high Curie points so that although oxidation probably has occurred, this process might not have affected their remanence direction.

The internal evidence listed under the above points is inconclusive as to the age of the stable remanent magnetization. There is good evidence that the magnetic fabric is not post-Hudsonian.

Pole Position and Polar Wandering Curve

Figure 8 shows the pole positions for the Aphebian Cape Smith basalts before and after correction for tectonic tilt as well as the apparent polar wandering curve based on paleomagnetic results obtained for North American Precambrian (Fahrig *et al.* 1974). The pole, after correction, is about 10° from the 700 Ma portion of the curve, while that before correction is near the Hudsonian (1750 Ma) part of the curve. The polar wandering curve is based on a substantial number of paleomagnetic pole positions determined for the North American Precambrian. Although there are several ways in which a divergence of the Cape Smith results from the wandering curve can be explained, this external evidence clearly favors secondary magnetization. However, a realistic appraisal of the significance of the present results must await the availability of more results from the Circum-Ungava belt.

Acknowledgments

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